A TRACE-CLASS OPERATOR WHOSE TRIANGULAR PROJECTION IS NOT TRACE CLASS

AGENT LANE 11

Source and Question

The source paper is Alice Barbara Tumpach, *Banach Poisson-Lie groups and Bruhat-Poisson structure of the restricted Grassmannian*, arXiv:1805.03292. In Remark 1.14 on page 9, after defining the triangular truncations T_- and T_{++}, the paper recalls the classical theorem that these truncations are unbounded on $L_\infty(\mathcal{H})$ and on $L_1(\mathcal{H})$, but bounded on $L_p(\mathcal{H})$ for $1 < p < \infty$ [1, Remark 1.14]. The same remark then asks for the existence and construction of a trace-class operator whose triangular projection is not trace class.

$$\langle m | D(A) n \rangle := \begin{cases} \quad 0 & \text{if} \quad n \neq m \end{cases} \tag{1.7}$$

Remark 1.14. The triangular truncations T_- and T_{++} are unbounded on $\mathrm{L}_\infty(\mathcal{H})$ and on $\mathrm{L}_1(\mathcal{H})$, but are bounded on $\mathrm{L}_p(\mathcal{H})$ for $1 < p < \infty$ (see [33], [28], [19] as well as Proposition 4.2 in [2] for the proof and more detail on the subject). See also [12] for an example of bounded operator whose triangular truncation is unbounded (Hilbert matrix). As far as we know the existence and construction of a trace class operator whose triangular projection is not trace class is an open problem. We refer the reader to [5] for related functional-analytic issues in the theory of Banach Lie groups.

Denote by $T_+ = T_{++} + D$ (resp. $T_{--} = T_- - D$) the linear transformation consisting in

Figure 1. Remark 1.14 on page 9 of the source paper, including the unboundedness statement and the trace-class triangular-projection question.

Proof Intuition

Unboundedness of triangular truncation on S_1 already contains the needed witness, but not in a single line. If every finite-dimensional triangular projection had norm bounded by one universal constant, then finite-rank operators would give a uniformly bounded dense-domain triangular truncation, and continuity would extend it to a bounded operator on all trace-class operators. This contradicts the source-cited unboundedness theorem.

Thus there are finite matrix blocks whose trace norm is tiny but whose triangular part has trace norm bounded below. Putting these blocks on disjoint consecutive diagonal intervals gives a trace-class direct sum. Triangular truncation acts independently on those diagonal blocks, so the first N block compressions of the triangular part have trace norm at least N. A trace-class operator cannot have uniformly unbounded finite compressions, so the triangular part is not trace class.

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Notation

Let $\mathcal{H} = \ell^2(\mathbb{N})$ with its usual ordered orthonormal basis $(e_j)_{j\geq 1}$. For a matrix $A = (a_{ij})$, write

$$\mathcal{T}(A) = (b_{ij}), \qquad b_{ij} = \begin{cases} a_{ij}, & i \leq j, \\ 0, & i > j. \end{cases}$$

This is the same entrywise triangular projection as the source paper's displayed definition of T_-, up to the harmless choice of indexing order. Write $S_1(\mathcal{H})$ for the trace-class operators on $\mathcal{H}$, with norm $\|A\|_1 = \mathrm{Tr}(|A|)$. If E is an n-dimensional initial coordinate subspace, $P_n : S_1(E) \to S_1(E)$ denotes the corresponding finite matrix triangular projection.

Main Result

Lemma 1 (finite-dimensional triangular norms are unbounded)**.** *The norms $\|P_n : S_1^n \to S_1^n\|$ are unbounded as $n \to \infty$.*

Proof. Suppose instead that $\|P_n\| \leq C$ for every n. Then for every finite-rank operator F supported on an initial coordinate subspace,

$$\|\mathcal{T}(F)\|_1 \leq C\|F\|_1.$$

Since such finite-rank operators are dense in $S_1(\mathcal{H})$, this inequality extends by continuity to a bounded linear map $R : S_1(\mathcal{H}) \to S_1(\mathcal{H})$. For fixed i, j, the coefficient functional $A \mapsto \langle Ae_j, e_i\rangle$ is continuous on $S_1(\mathcal{H})$. Hence approximating $A \in S_1(\mathcal{H})$ by finite-rank initial compressions shows that the matrix coefficients of RA are exactly those of the entrywise triangular truncation $\mathcal{T}(A)$.

Thus triangular truncation would be a bounded operator on $S_1(\mathcal{H})$. This contradicts the classical unboundedness theorem recalled in Remark 1.14 of the source paper [1], which cites Macaev, Kwapien–Pelczynski, Gohberg–Krein, and Arazy for this fact [5, 4, 3, 2]. □

Theorem 1. *There exists $A \in S_1(\mathcal{H})$ such that the entrywise triangular projection $\mathcal{T}(A)$ is not trace class. Moreover A can be chosen as a block diagonal direct sum of finite-rank operators on consecutive coordinate blocks.*

Proof. By Lemma 1, for each $k \geq 1$ there are an integer n_k and a matrix $C_k \in S_1^{n_k}$ such that

$$\|C_k\|_1 = 1, \qquad \|P_{n_k} C_k\|_1 \geq 2^k.$$

Set $B_k = 2^{-k} C_k$. Then

$$\|B_k\|_1 = 2^{-k}, \qquad \|P_{n_k} B_k\|_1 \geq 1.$$

Choose pairwise disjoint consecutive intervals

$$I_k = \{m_k + 1, \ldots, m_k + n_k\} \subset \mathbb{N}$$

and order-preserving unitaries $U_k : \mathbb{C}^{n_k} \to \ell^2(I_k)$. Define

$$A_k = U_k B_k U_k^*, \qquad A = \bigoplus_{k=1}^{\infty} A_k$$

with respect to the orthogonal decomposition $\mathcal{H} = \bigoplus_k \ell^2(I_k)$ plus any unused coordinates. Since

$$\sum_{k=1}^{\infty} \|A_k\|_1 = \sum_{k=1}^{\infty} 2^{-k} < \infty,$$

the block diagonal operator A is trace class.

Because the blocks are consecutive and all off-block matrix coefficients of A vanish, triangular truncation preserves the block diagonal form: on the block I_k, the compression of $\mathcal{T}(A)$ is

$$U_k(P_{n_k}B_k)U_k^*.$$

Let Q_N be the orthogonal projection onto the sum of the first N blocks. Then

$$\begin{aligned}\|Q_N\mathcal{T}(A)Q_N\|_1 &= \sum_{k=1}^{N}\|U_k(P_{n_k}B_k)U_k^*\|_1\\ &= \sum_{k=1}^{N}\|P_{n_k}B_k\|_1 \geq N.\end{aligned}$$

If $\mathcal{T}(A)$ were trace class, then $\|Q_N\mathcal{T}(A)Q_N\|_1 \leq \|\mathcal{T}(A)\|_1$ for every N, an impossibility. Therefore $\mathcal{T}(A) \notin S_1(\mathcal{H})$. □

Consequence for the Source Question

Theorem 1 supplies the trace-class operator requested in Remark 1.14. The construction is canonical once one selects a sequence of finite-dimensional matrices witnessing the unboundedness of triangular projection on S_1^n. The same paper later uses unboundedness of triangular truncation to choose a sequence $K_n \in L_1$ with bounded trace norm and triangular truncations of arbitrarily large trace norm [1, Proposition 7.1 proof]; the present argument upgrades this "unbounded sequence" phenomenon to one single trace-class operator by the block-direct-sum device.

Verification Notes

No numerical verification is needed for the proof. The two functional-analytic checks are:

(1) finite-dimensional triangular norms must be unbounded, otherwise the dense finite-rank truncation extends boundedly to all S_1;
(2) for a trace-class operator C, finite-rank compressions satisfy $\|QCQ\|_1 \leq \|C\|_1$, while the constructed triangular truncation has compressions with norms tending to infinity.

The script in `code/render_open_problem_crop.py` only regenerates the source-page crop.

Novelty Check

Before promotion, the run checked the lightweight registry, solution, attempt, and proof-gap indexes for arXiv:1805.03292 and the phrases "triangular projection", "triangular truncation", "trace class operator", "not trace class", "Macaev", "Kwapien", "Pelczynski", "Gohberg", "Krein", and "Arazy". No duplicate packet was found. A local parsed-corpus search found the source question, the same paper's later use of an unbounded sequence of trace-class witnesses, and related later arXiv records on triangular projection for $0 < p < 1$, but no separate answer to the S_1 single-operator question. Web searches for exact phrases including "trace class operator triangular projection not trace class", "triangular truncation trace class not trace class", and "Tumpach triangular projection trace class" did not reveal a later paper explicitly answering Remark 1.14.

Limitations

This packet answers only the trace-class triangular-projection question in Remark 1.14. It does not address the separate broad problem, also present in the source paper, about integrating Banach Lie bialgebras into Banach Poisson-Lie groups. The construction is explicit at the operator-assembly level but depends on choosing finite-dimensional matrices from the classical unboundedness theorem for triangular projections on S_1^n.

Human-Review Recommendation

Review as a likely valid full answer to the trace-class triangular-projection open problem in Remark 1.14. The main point to verify is the passage from source-cited unboundedness on $S_1(\mathcal{H})$ to unbounded finite-dimensional triangular projection norms in Lemma 1.

References

[1] A. B. Tumpach, *Banach Poisson-Lie groups and Bruhat-Poisson structure of the restricted Grassmannian*, arXiv:1805.03292; Comm. Math. Phys. 373 (2020), 795–858.

[2] J. Arazy, *Some remarks on interpolation theorems and the boundness of the triangular projection in unitary matrix spaces*, Integral Equations Operator Theory 1 (1978), 453–495.

[3] I. C. Gohberg and M. G. Krein, *Theory and Applications of Volterra Operators on Hilbert Space*, Translations of Mathematical Monographs, vol. 24, American Mathematical Society, 1970.

[4] S. Kwapien and A. Pelczynski, *The main triangle projection in matrix spaces and its applications*, Studia Math. 34 (1970), 43–68.

[5] V. I. Macaev, *Volterra operators produced by perturbation of self-adjoint operators*, Soviet Math. Dokl. 2 (1961), 1013–1016.